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FREE GUIDE · 2026 EDITION

Genotype, Marriage and Your Options

A clear, practical guide to sickle cell for couples, before marriage and before starting a family.

By AwaDoc Research Team

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WELCOME

Before you begin

If you're reading this, you've probably just found out your genotype, or your partner's, and you have questions. Maybe you're both AS. Maybe a parent has told you that you can't marry the person you love. Maybe you're married already and planning your first child. Whatever brought you here, you're in the right place.

We wrote this guide because couples ask us the same questions again and again, and too often the answers they hear are frightening, outdated or simply wrong. Your genotype matters. It tells you real things about your future children. But it should inform your choices, not make them for you.

In the pages that follow, we explain what your results mean, the true odds for your children, and every option open to you today, including some that were not available in Nigeria a generation ago. We have kept the language plain and the facts honest, including the limits of what medicine can do.

> **How to use this guide**
> Read it together with your partner if you can. Chapters 1 to 3 cover the basics. Chapters 4 to 7 explain your options and what they involve. Chapters 8 to 10 help you choose a clinic, talk to the people you love, and take your next step. A short glossary at the end explains medical words.

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CHAPTER 1

Your genotype, explained

Your genotype is simply the pair of haemoglobin genes you carry, one from your mother and one from your father. Understanding those two letters is the first step to understanding everything else in this guide.

Haemoglobin is the protein inside red blood cells that carries oxygen around your body. The gene that makes it comes in a few versions. The letters in your genotype tell you which versions you have.

The three letters you'll see

- **A** is the usual haemoglobin gene. It makes red blood cells that are soft, round and flexible.
- **S** is the sickle gene. Red blood cells made with two S genes can become stiff and curved, like a sickle, especially when oxygen is low. These cells can block small blood vessels and break down early.
- **C** is another variant. On its own it is mild, but paired with an S gene it causes a form of sickle cell disease.

Genotype	What it means	Effect on health
AA	Two usual genes	No sickle cell trait and no sickle cell disease
AS	One usual gene and one sickle gene	Carrier, also called sickle cell trait. Usually completely healthy
AC	One usual gene and one C gene	Carrier of haemoglobin C. Usually completely healthy
SS	Two sickle genes	Sickle cell anaemia, the most common and often the most severe form of the disease
SC	One sickle gene and one C gene	Sickle cell disease, often milder than SS but still serious
CC	Two C genes	Haemoglobin C disease, usually causing only mild anaemia

Carrier or disease: the difference that matters

A carrier has one sickle gene and one usual gene. The usual gene does enough work to keep the carrier healthy, so most carriers never have symptoms and live full, normal lives. Sickle cell disease happens only when a person inherits a sickle gene from *both* parents, or a sickle gene from one parent and a C gene from the other.

> **Key point**
> Being AS is very common. About 1 in 4 Nigerian adults carries the sickle gene. There is nothing wrong with you, and nothing to be ashamed of. What matters for your children is the combination of genes you and your partner carry together.

CHAPTER 2

How sickle cell is passed on

Every child receives one haemoglobin gene from each parent, chosen by chance. Once you know both genotypes, you can work out the exact odds for every pregnancy.

Think of each parent as holding two cards. For each pregnancy, each parent passes on one of their two cards, at random. The child's genotype is the pair of cards they receive. Because the choice is random every time, the odds below apply to **each pregnancy separately**.

Couple	Chance for each pregnancy	Risk of sickle cell disease	Can embryo testing help?
AA + any	No child can have sickle cell disease	0%	Not needed
AS + AS	AA 25%, AS 50%, SS 25%	25%	Yes
AS + AC	AA 25%, AS 25%, AC 25%, SC 25%	25%	Yes
AS + SS	AS 50%, SS 50%	50%	Yes
AS + SC	AS 25%, AC 25%, SS 25%, SC 25%	50%	Yes
AC + SS	AS 50%, SC 50%	50%	Yes
AC + AC	AA 25%, AC 50%, CC 25%	0% for sickle cell	Not needed for sickle cell
SS + SS	SS 100%	100%	No. Donor eggs or sperm are the option

What "25%" really means

A 25% risk does **not** mean that one in every four of your children will have sickle cell disease. It means that each pregnancy carries a one in four chance, no matter what happened before. An AS + AS couple could have three healthy children in a row, or two affected children in a row. Previous pregnancies don't change the odds for the next one.

This is why couples who already have an affected child sometimes feel surprised when a second child is affected too, and why couples with several healthy children should not assume the risk has passed.

> **Remember**
> If either partner is AA, none of your children can have sickle cell disease. At most, a child could be a carrier like the other parent, which does not cause illness.

CHAPTER 3

Getting tested properly

Every decision in this guide depends on accurate results for both partners. Testing is quick and affordable, but it has to be done the right way.

When to test

The best time is early, before a relationship becomes serious and certainly before marriage or trying for a baby. Both partners need a result. One person's genotype on its own tells you nothing about your future children.

Which test to ask for

Ask for **haemoglobin electrophoresis** or **HPLC** (high performance liquid chromatography). These tests show exactly which types of haemoglobin you have. A routine blood count, or a quick test that only says "positive" or "negative", is not enough.

Why you may need to test again

Genotype results are wrong more often than people realise. Samples get mixed up, results are copied by hand, and some quick tests miss the C gene completely. Families have made life changing decisions based on a single wrong result. If anything about your result feels uncertain, test again at an accredited laboratory.

> **Warning signs that your result may be wrong**
>
> - You have two results that don't agree with each other
> - Your result came from a quick test at an event, a church or an outreach
> - You were told you are AA, but a parent or sibling has sickle cell disease
> - Your result says only "positive" or "negative", with no letters
> - Your result is many years old and you don't have the original report

A genotype confirmation test costs ₦30,000. The test is carried out by our clinical partner, IMSA, and you can book it through AwaDoc. If you are unsure of your result, choose "Not sure" when you book, and we will schedule the test with IMSA for you.

CHAPTER 4

Your options as a carrier couple

Learning that you are both carriers is not the end of your plans. You have several paths open to you, and there is no single right answer. The best choice depends on your values, your faith, your health and your budget.

1. Conceive naturally and prepare

Many couples choose this, particularly AS + AS couples, where each child has a three in four chance of not having the disease. Your baby can be tested soon after birth. Early diagnosis and good care, including protection from infection, make a real difference to how children with sickle cell disease grow and thrive.

2. Test during pregnancy

A doctor can find out the baby's genotype during pregnancy. Chorionic villus sampling (CVS) is usually done from about 11 to 14 weeks, and amniocentesis from about 15 weeks. Both carry a small risk of miscarriage, under 1 in 100. Knowing early helps families prepare. Some couples also consider ending an affected pregnancy, but Nigerian law on this is very restrictive, so speak to your doctor about what is possible where you live.

3. IVF with embryo testing (PGT-M)

Embryos are created through IVF and tested before a pregnancy begins. Only embryos without sickle cell disease are placed in the womb. This avoids difficult decisions during pregnancy, but it requires IVF and is the most expensive option. Chapters 5 to 7 explain it in full.

4. Donor eggs or sperm

Using eggs or sperm from a donor who is AA means no child can have sickle cell disease. This is the main option for SS + SS couples, for whom embryo testing cannot help.

5. Adoption or fostering

Some couples build or grow their family this way, on its own or alongside other options.

6. Choosing your own path together

Some couples decide not to have children together. Others marry and accept the risk. Both are valid. Your genotype should inform your plans, not decide your relationship.

> **Talk it through first**
> Genetic counselling gives you time with a doctor who explains each option, answers your questions honestly and helps you weigh what matters most to you, without pressure.

CHAPTER 5

IVF with embryo testing, step by step

PGT-M stands for preimplantation genetic testing for monogenic conditions. It lets carrier couples begin a pregnancy with an embryo that does not have sickle cell disease.

It is important to understand what PGT-M does and does not do. It **selects** embryos. It does not change, repair or edit any genes. IVF usually produces several embryos, and testing simply tells you which of them are free of the disease.

Step	What happens	How long
1. Counselling	An AwaDoc doctor reviews both genotype results, explains your chances and checks that this option suits you.	1 session, online
2. Family DNA workup	At IMSA, you both give blood or cheek swab samples, ideally with an affected relative's too. The laboratory builds a test designed for your family.	3 to 6 weeks
3. Ovarian stimulation	Daily hormone injections help several eggs mature, with scans to track their growth.	About 10 to 14 days
4. Egg collection	Eggs are collected under sedation and fertilised in the laboratory.	1 day
5. Embryo testing	On day 5 or 6, a few cells are taken from the outer layer of each embryo and tested. Embryos are frozen while results come back.	1 to 3 weeks
6. Embryo transfer	An embryo without sickle cell disease is placed in the womb, usually in a later cycle.	1 day
7. Confirmation	A pregnancy test about two weeks later. A confirmatory test during pregnancy is still advised.	From 2 weeks

Most couples go from counselling to embryo transfer in about 3 to 4 months. AwaDoc handles your counselling, booking and follow up. Every test and procedure, from DNA samples to embryo transfer, is carried out by our clinical partner, the Institute of Medical Sciences Africa (IMSA) in Abuja, usually across three visits. Couples living abroad can start online and travel only when they need to be at the clinic.

Who it can help

AS + AS, AS + AC, AS + SS, AS + SC and AC + SS couples. It cannot help SS + SS couples, because every embryo would be affected.

CHAPTER 6

What the evidence says

Embryo testing for sickle cell is not new or experimental. It has been used for over 25 years, and the results are well documented.

Success rates

The largest study of couples using PGT-M specifically for sickle cell followed 60 couples at a London hospital between 2012 and 2017. **63% of couples had a baby**, and 59% of tested embryos were free of the disease and suitable for transfer. There were no misdiagnoses.

Across a large European registry covering many centres and many conditions, the clinical pregnancy rate was about 35% for each embryo transfer. Rates for each couple are higher than rates for each transfer, because many couples have more than one transfer.

Accuracy

Accuracy has improved a great deal. Early methods had a misdiagnosis rate of about 0.4%. A 2025 study of more than 9,000 cases using a modern method found no misdiagnoses among the cases that could be checked. No test is perfect, which is why a confirmatory test during pregnancy is still recommended.

The Nigerian experience

Nigerian couples have used embryo testing for sickle cell since at least 2012, when early cases relied on laboratories abroad. In 2020, a team in Abuja led by Dr Ibrahim Wada reported a healthy AA baby boy born after embryos from an AS + AS couple were tested within Nigeria. At the same Abuja centre, sickle cell was the reason for over 94% of embryo testing cycles across five years.

What age means for success

A woman's age is the strongest single factor in IVF success. Results are best under 35 and fall steadily after that. This is why online treatment booking with AwaDoc is for couples where the woman is 35 or under. Older couples can still book counselling to discuss their options.

CHAPTER 7

Costs and planning

IVF with embryo testing is a significant investment. Knowing the full cost upfront helps you plan with confidence and avoid surprises.

Service	Price
Virtual counselling, from any country	₦50,000
In person counselling	₦80,000
Genotype confirmation test	₦30,000
IVF cycle	₦2,900,000 estimate
Fertility medication	₦1,600,000 estimate
Embryo testing (PGT-M)	₦2,500,000 estimate
Frozen embryo transfer	₦500,000 estimate
Full treatment package	**₦7,500,000**

All tests and procedures are carried out by IMSA. AwaDoc books them for you and takes one payment for everything. Procedure prices are estimates until IMSA confirms them at your assessment. Payment is made in full, in one payment, by card, bank transfer or mobile money.

What isn't included

- Flights, visas and travel insurance
- Accommodation and local transport in Abuja
- Meals and personal costs
- Extra IVF cycles, if needed
- Pregnancy and birth care

Refunds

Cancel counselling at least 24 hours before your session to reschedule once or receive a full refund. If IMSA finds at your assessment that treatment isn't suitable for you, procedure payments are refunded in full.

Help with the cost

You don't have to fund treatment alone. Some couples are supported by family, employers or donors. Tell us during counselling and we'll talk through the options with you.

CHAPTER 8

Questions to ask any clinic

Nigeria does not yet have a national law regulating fertility treatment, so the quality of care varies. A good clinic will answer every question below clearly, and in writing.

- [ ] Is embryo testing done in your own laboratory, or are samples sent abroad?
- [ ] Which testing method do you use, and how do you check its accuracy?
- [ ] How many sickle cell embryo testing cycles have you done, and how many led to a baby?
- [ ] What are your success rates for women of my age?
- [ ] What happens if no embryo is suitable for transfer?
- [ ] What happens to unaffected embryos we don't use? Can we freeze, donate or discard them?
- [ ] Exactly what does the price include, and what costs extra?
- [ ] What is your refund policy if treatment is stopped?
- [ ] Do you offer sex selection? A clinic focused on preventing disease should decline non medical requests.
- [ ] Who can we contact with questions between visits?

Bring this list to your consultation, wherever you choose to be treated. It is your right to understand exactly what you are paying for.

CHAPTER 9

Talking to your partner, family and faith leaders

Genotype conversations can feel heavy, but they go better when you plan them. You are making a decision about your future family, not confessing a problem.

With your partner

- Share your results early, ideally before talk of marriage becomes serious.
- Say how you feel, and listen without rushing to a solution.
- Learn the facts together. Book counselling as a couple so you both hear the same information.
- Agree that nobody is to blame. Genes are inherited, never chosen.

With your families

- Decide together what you will share, and who will speak to which side of the family.
- Lead with the plan, not only the problem. For example: "We're both AS, and we've learned there are safe ways to have children free of sickle cell."
- Expect strong feelings. Many parents grew up when AS + AS simply meant "don't marry". Sharing this guide can help.
- Remember that the final decision belongs to the two of you.

With your faith community

- Many religious leaders now encourage genotype testing before marriage.
- If IVF or embryo testing raises questions of faith for you, speak with a trusted leader early.
- Bring the facts: embryos are tested and selected, nothing is edited, and you can ask the clinic what happens to embryos you don't use.

> **Look after yourselves**
> Worry, sadness and frustration are normal reactions. If they don't ease, or begin to affect your daily life, please speak to a doctor or counsellor. Asking for support is a sign of strength.

CHAPTER 10

Your next steps

You don't need to decide everything today. Start with the facts, take one step at a time, and make the choice that feels right for your family.

1. **Know both genotypes for sure**
   If either result is old or uncertain, book a ₦30,000 confirmation test through AwaDoc. IMSA carries out the test.

2. **Check your odds**
   Find your couple in the table in Chapter 2.

3. **Talk it through with a doctor**
   Book virtual counselling for ₦50,000 from any country, or in person counselling for ₦80,000 in Abuja.

4. **Decide what's right for you**
   With the facts in hand, and with no pressure from anyone.

Get in touch

Book online, ask Noura on the website at any time, or email us. We usually reply within 2 working days.

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REFERENCE

Glossary

Amniocentesis	A test during pregnancy, from about 15 weeks, that checks a sample of the fluid around the baby.
Carrier	A person with one usual gene and one variant gene, such as AS. Carriers are usually healthy.
CVS	Chorionic villus sampling. A test during pregnancy, from about 11 weeks, that checks a tiny sample of the placenta.
Embryo	The earliest stage of development after an egg is fertilised.
Embryo transfer	Placing an embryo in the womb so that a pregnancy can begin.
Genotype	The pair of genes you carry for a particular trait, such as AA or AS.
Haemoglobin	The protein in red blood cells that carries oxygen.
Haemoglobin electrophoresis	A laboratory test that identifies which types of haemoglobin you have.
HPLC	High performance liquid chromatography. Another accurate test for haemoglobin types.
IVF	In vitro fertilisation. Eggs are fertilised with sperm in a laboratory, and an embryo is then placed in the womb.
PGT-M	Preimplantation genetic testing for monogenic conditions. Testing embryos for a single inherited condition such as sickle cell disease.
Sickle cell disease	An inherited blood condition caused by inheriting two sickle genes (SS), or a sickle gene and a C gene (SC).
Sickle cell trait	Carrying one sickle gene (AS). Usually causes no symptoms.

Sources

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Full references are available at genomics.awadoc.com/evidence.

Disclaimer. This guide provides general information and is not medical advice. Please discuss your own situation with a qualified doctor. AwaDoc provides genotype counselling, booking and follow up, and refers couples for treatment. All tests and procedures are carried out by our clinical partner, the Institute of Medical Sciences Africa (IMSA).

Your genotype shouldn't decide who you marry.

AwaDoc, in partnership with IMSA, helps couples who carry the sickle cell gene understand their options and have children free of the disease, with expert care that is private, compassionate and accessible from anywhere.

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